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Dear Editors,

On behalf of all co-authors (Rodrigo Almeida de Oliveira, Juliano de Paulo Ribeiro, and Prof. Edson Emilio Scalabrin), I am pleased to submit our manuscript titled **"Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping of Stored Family Codes"** for consideration in *Information and Software Technology*.

Novelty and contribution. This manuscript reports a systematic mapping study at the intersection of process mining (PM), stochastic modeling, and machine learning applied to software process forecasting. The contribution is threefold:

1. **Methodological:** an LLM screened the records (Claude Haiku 4.5 as primary screener, Claude Sonnet 4.6 as verifier). On the parseable cross-model sample, binary Cohen's κ is 0.695 at title/abstract (420 of 468 papers; 48 verifier responses had no category because the abstract was absent) and 0.694 at the second pass ($n = 177$). That agreement is not a human validation. The pre-registered check is a two-rater sample. Recall on the working-set sample is 0.630 (29 of 46). It is not a full-text Recall: a PDF was read for 0.5% of the 886 records. The round-1 FT row (Recall 0.246; $\kappa = 0.122$) is withdrawn. Auxiliary Recall after the stratum re-screen is 0.725 (235 of 324). Both are below the pre-registered bar of 0.90. The two reviewers agreed with 92% of the suggested labels on the remaining records. The re-screen sheets (4,005 rows) match the suggestion on every row, so that share is not marked there. Rebuilt from the title or abstract, it is 32 of the 338 coded includes (9.5%), and those 32 are removed.
2. **Empirical:** the corpus of this revision is 456 studies: 306 that remain after 32 coded studies are removed from the list of 338, plus the 150 added studies (ADD-150) that remain after an eligibility pass removed 18 of 168. The family codes are reported on WS-151, the 151 working-set studies inside the 306, a cut disjoint from ADD-150. One study in the 380 with at least one family mark (PRIMAD) matches process mining, a stochastic model, and forecasting. Seventy-six added studies, read from the title alone, are outside that fraction. F2 and F5 are withdrawn. F1, F3, and F4 are stored codes and are not findings. The agenda is not inferred from them.
3. **Synthetic:** the SPMF taxonomy and a four-direction research agenda (RA1–RA4), proposed by the authors and not inferred from F1–F5. PATHCAST is future work. It is not a result of this mapping.

Fit with IST scope. The manuscript squarely fits IST's scope on empirical software engineering methods, mining software repositories, and process modeling. It complements the journal's

growing body of evidence-based secondary studies while contributing to the methodological literature on LLM-assisted reviews.

Reproducibility. A complete replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600), including all search strings, screening prompts, JSON schemas, decision CSVs at every pipeline stage, the 456-study included list (`included_studies_506.csv`), the cross-model κ scripts, and the structured extraction tables.

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Prior submission and conflicts. The manuscript is original work, has not been published elsewhere, and is not under consideration by any other journal. All authors declare no competing financial interests. The use of generative AI tools in the research is declared in the manuscript per Elsevier policy. AI tools are *not* listed as authors and bear no responsibility for editorial judgements.

Suggested reviewers.

- **Wil M. P. van der Aalst** — RWTH Aachen University, foundational PM researcher (`wvdaalst@pads.rwth-aachen.de`)
- **Maria Teresa Baldassarre** — University of Bari, empirical SE and secondary studies
- **Eugene Syriani** — University of Montreal, LLM-assisted SLR methodology
- **Tore Dybå** — SINTEF / University of Oslo, evidence-based SE

Reviewers to avoid. None on basis of co-authorship within the past five years. Specific exclusions: thesis advisors and committee members at PUCPR.

Sincerely,

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Paraná (PUCPR)